

# Coastal observations: reproducible data management

Document: Science Europe metadata follow-up experiment

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## Research overview

|                       |                                            |
|-----------------------|--------------------------------------------|
| <b>Project Number</b> | COAST-2026                                 |
| <b>Start date</b>     | 2026-01-01                                 |
| <b>End date</b>       | 2027-12-31                                 |
| <b>Funding</b>        | Funding information has not been provided. |

Synthetic scenario for testing all fifteen output questions. New coastal temperature measurements will be compared with two open reference datasets. This is not an approved or complete DMP.

## People and responsibilities

- **Example research team**

[research@example.org](mailto:research@example.org)

Roles: Contact Person, Creator of DMP, Data Steward

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Knowledge model: Common DSW Knowledge Model, 2.7.0 (dsw:root:2.7.0)

Generated with [Data Stewardship Wizard](#).

## Section 1: Data description and collection or re-use of existing data

1. How will new data be collected or produced and/or how will existing data be re-used?

### Instrument datasets

- **Coastal temperature measurements**

For this dataset, we are using the following instruments:

- **Temperature logger**
  - Record temperature every ten minutes; retain UTC timestamps and calibration records.

Quality control measures for **Coastal temperature measurements**: measurement calibration, data entry validation.

### Re-used datasets

#### Open coastal observations — station 1

Source: <https://example.org/datasets/coastal-observations/station-1/version-2026>

Compare seasonal changes in water temperature.

- Retain the original timestamps.
- Document any excluded observations.

|         |        |
|---------|--------|
| Version | 2026.1 |
|---------|--------|

If a new version becomes available during the project, the current version will continue to be used.

#### Open coastal observations — station 2

Source: <https://example.org/datasets/coastal-observations/station-2/version-2026>

Compare seasonal changes in water temperature.

- Retain the original timestamps.
- Document any excluded observations.

|         |        |
|---------|--------|
| Version | 2026.1 |
|---------|--------|

If a new version becomes available during the project, the current version will continue to be used.

### Shared arrangements

The following arrangements apply to all reference datasets selected for re-use above.

We will retain a copy of this dataset and provide it alongside the research results. This dataset is freely available with an obligation to cite the source. The data can be used directly without format conversion.

## Provenance

Record source versions and processing scripts in the project repository.

| Record          | Retention |
|-----------------|-----------|
| Processing log  | 10 years  |
| Source checksum | 10 years  |

2. What data (for example the kinds, formats, and volumes) will be collected or produced?

### Instrument datasets

Data collection for **Coastal temperature measurements**: This dataset will be collected by experts in the project, with our own equipment. The equipment is very well described and known.

### Data formats and types

Data format: **CSV (UTF-8)**. It is a standardized format suitable for long-term archiving. The estimated data volume is 120 GB.

## Section 2: Documentation and data quality

3. What metadata and documentation (for example the methodology of data collection and way of organising data) will accompany the data?

We will be documenting the data with Dublin Core metadata standard. Metadata will be openly available.

### Data storage and file conventions

Information not provided: the estimated storage space for all project data and software, including temporary storage (GB).

We will use a filesystem with files and folders. We document how we manage file versioning for files and folders.

File naming conventions:

Name files station\_YYYYMMDD.csv and document corrections in CHANGELOG.md.

4. What data quality control measures will be used?

Quality control measures for **Coastal temperature measurements**: measurement calibration, data entry validation.

## Section 3: Storage and backup during research process

5. How will data and metadata be stored and backed up during the research process?

There is a shared workspace used during the project for working with data. Storage needs will be the same during the whole project. All essential data will also be stored elsewhere to prevent a total loss of data. We will make automated backups of all data stored outside the workspace.

The fields mapped by this template are insufficient to establish the following operational details:

- Exact working-storage and backup locations are not established by these mapped answers.
- An actual backup schedule is not established by these mapped answers; a need for frequent backups is not a schedule.

6. How will data security and protection of sensitive data be taken care of during the research?

Project members will not carry data with them (e.g. on laptops, USB sticks, or other external media). All project web services are accessible via secure HTTP (<https://...>). Project members have been instructed about both generic and specific risks to the project.

## Section 4: Legal and ethical requirements, codes of conduct

7. If personal data are processed, how will compliance with legislation on personal data and on security be ensured?

We will not collect any data related to individuals.

8. How will other legal issues, such as intellectual property rights and ownership, be managed? What legislation is applicable?

For the reference and non-reference data sets that we reuse, conditions are as follows:

- Open coastal observations — station 1  
It is freely available with obligation to quote the source (e.g. CC-BY).
- Open coastal observations — station 2  
It is freely available with obligation to quote the source (e.g. CC-BY).

9. What ethical issues and codes of conduct are there, and how will they be taken into account?

### Ethical approvals

The questionnaire records the following ethical approval arrangements:

- **Coastal observations: reproducible data management**  
According to the questionnaire, this project does not require ethical approval.

## Data we produce

For the data we produce, the ethical aspects are as follows:

- **Quality-checked coastal observations**
  - It does not contain personal data.
  - It does not contain sensitive data.

## Data we collect

We will not collect any data related to individuals, i.e. "personal data".

The data collection is not subject to ethical legislation.

## Section 5: Data sharing and long-term preservation

10. How and when will data be shared? Are there possible restrictions to data sharing or embargo reasons?

### Quality-checked coastal observations

This dataset will be published.

Open access: this distribution will be shared with anyone. This distribution will be deposited in an institutional repository. Licence start date: 2027-12-31. Freely available with obligation to quote the source (e.g. CC-BY).

11. How will data for preservation be selected, and where will data be preserved long-term (for example a data repository or archive)?

### Quality-checked coastal observations

This dataset will be published. Retention period (prepaid): 10 years. The metadata will be available even when the data no longer exists.

Preservation destinations by distribution:

- Our institutional repository.

We have budgeted for the costs in the project to pay charges of the used repositories. We have a reserved budget for the time and effort it will take to prepare the data for publication.

Preservation selection still requires review: publication decisions and archive renewal criteria do not by themselves explain which datasets and versions must be retained or destroyed.

12. What methods or software tools are needed to access and use data?

### Quality-checked coastal observations

Software required to use this dataset:

- **CoastView 1.0 (synthetic test tool)**  
Available at: <https://example.org/coastview/1.0>

13. How will the application of a unique and persistent identifier (such as a Digital Object Identifier (DOI)) to each data set be ensured?

### Quality-checked coastal observations

#### Institutional repository

The repository will assign the persistent identifier. The repository will make sure the persistent identifier can be resolved to a digital object.

#### Possible re-use of created data in other contexts

- **Coastal temperature measurements**

Other researchers in this field will be interested in re-using this data.

## Section 6: Data management responsibilities and resources

14. Who (for example role, position, and institution) will be responsible for data management (i.e. the data steward)?

Example research team is responsible for the management and proficiency of data including data processing, data policies, data guidelines, and data availability.

Example research team is responsible for implementing the DMP, and ensuring it is reviewed and revised.

15. What resources (for example financial and time) will be dedicated to data management and ensuring that data will be FAIR (Findable, Accessible, Interoperable, Re-usable)?

Additional specialist expertise is required. Existing staff will receive training.

Staff will complete training before data publication.

- Metadata quality checks
- Version control and provenance

We do not require any hardware or software in addition to what is usually available in the institute.

The selected repositories charge for their services.

### Data-management budget

| Resource and purpose                                                                                                                             | Budget   | Funding source               |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------|
| <b>Data preparation resource 1</b><br>Prepare metadata and validate deposit files.<br>This resource supports findability and management of data. | 5000 TWD | Institutional research funds |
| <b>Data preparation resource 2</b><br>Prepare metadata and validate deposit files.<br>This resource supports findability and management of data. | 0 TWD    | Institutional research funds |